

Data Analytics (DA)

Module-End -4

Module End Assignment: Python

“Customer Support Ticket Analyzer”

Assignment Task Overview

Project Title: Customer Support Ticket Analyser

Problem Statement

Customer support teams handle numerous service tickets daily. Analysing support tickets helps identify common issues, customer sentiment, support quality, and areas for improvement.

In this project, you will build a **Python-based Ticket Analysis System** that stores, cleans, analyses, and extracts insights from customer support tickets.

Step 1: Preloaded Tickets

Begin your program with the following dictionary of lists containing 10 customer tickets:

```
ticket_data = { 'Ticket_No': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],

'Customer_Name': [ 'Ravi', 'Meera', 'Sam', 'Anu', 'Rakesh', 'Divya', 'Arjun', 'Kiran', 'Leela', 'Nisha'

],

'Issue_Description': [

' Internet not working!!! ', 'slow response, very poor service ',

'GREAT support! issue resolved.', 'okay... need help ', 'not BAD but slow', 'Excellent guidance, Very Helpful!',

'good support and good behaviour!', 'Poor handling of technical issue', 'Satisfied. Could be better.', 'Good service...

quick response.'

],

'Priority': ['High', 'Low', 'High', 'Medium', 'Low', 'High', 'Medium', 'High', 'Low', 'Medium']}

print(ticket_data)
```

- **Task:** Print the initial ticket data in a readable format.

Step 2: Add More Tickets

- ❖ Ask the user: **How many new tickets do you want to add?**
- ❖ For each new ticket, collect:
 - Customer Name
 - Issue Description
 - Priority (High / Medium / Low)

Requirements:

- Ticket numbers must auto-increment starting from 11.
- Validate priority (accept only High, Medium, Low).
- Append new data to `ticket_data`.

Step 3: Text Cleaning for Issue Descriptions

- ❖ Clean **all issue descriptions** by applying:
 - Remove punctuation (.,!?-)
 - Convert multiple spaces → single space
 - Remove leading/trailing spaces
 - Convert text to lowercase
 - Replace slang/shorthand (example: "ok" → "okay")
- ❖ **Hints:**
 - Use `.replace()`, `.split()`, `' '.join()`, `.strip()`, `.lower()`.

Step 4: Keyword-Based Issue Insights

Create a function:

`def count_tickets_with_word(word):`

Function Requirements:

- Case-insensitive search
- Returns how many ticket descriptions contain the given word

Use this function to print:

- Number of tickets containing "poor"
- Number of tickets containing "good"
- Number of tickets containing "slow"
- Number of tickets containing "excellent"

Step 5: Final Summary & Insights

Produce detailed analytics:

1. Display Final Cleaned `ticket_data`

Nicely formatted dictionary-of-lists output.

2. Priority Analysis

Compute:

- Number of **High-priority** tickets
- Number of **Medium priority** tickets
- Number of **Low-Priority** tickets

3. Find the Ticket With the Longest Issue Description

Based on **word count**, print:

- Ticket number
- Customer name
- Cleaned issue text
- Word count

4. Extract Unique Words Used

Generate a **set** of all unique words across all issue descriptions.

Show:

- Count of unique words
- The word list (sorted)

Deliverables:

- Complete the assignment in Google Colab or Jupyter Notebook.
- For Google Colab, share the notebook link with "View" access. If using Jupyter, upload the notebook to Google Drive and ensure "View" access is enabled.
- Clear one-page summary with findings supported by analysis
- Put both the Google Colab notebook and the summary document in a Google Drive folder and share that folder with view access

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Expected Proficiency
Data Loading & Organization	Python Basics	Lists, Dictionaries, Appending Data	Beginner
Data Cleaning & Transformation	Text Processing	Lowercasing, punctuation removal, spacing	Intermediate
Keyword & Text Analysis	Logical/Analytical Skills	Word search, counting, string operations	Intermediate
Summary Statistics	Data Analysis	priority distribution, longest text, unique words	Intermediate
Data Structuring & Sorting	Data Handling	Sorting with filtering, restructuring	Intermediate
Reporting & Insights	Communication	Clear summary of findings & patterns	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill/Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Credit (Rounded Range)
Data Loading & Setup	3	Data correctly loaded, structured, and appended	Minor issues in loading or formatting	1–2
Data Cleaning & Text Processing	5	All cleaning steps completed accurately (lowercase, punctuation removal, spacing, slang fix)	Some steps correct, but partially incomplete or inconsistent	3–4
Functions, Loops & Logic	4	All required functions work correctly (word count, loops, conditionals)	Functions partially correct or missing edge cases	2–3
Summary Statistics	4	Accurate average rating, longest feedback, unique word extraction	Most calculations correct but incomplete	2–3
Sorting & Structuring Output	3	Correct ranking/sorting of tickets and structured display	Partially sorted or formatting errors	1–2
Output Presentation	4	Clear, readable outputs in notebook (tables/lists well formatted)	Outputs present but not clearly formatted	2–3
Reporting & Insights	2	Clear one-page summary with findings supported by analysis	Summary incomplete or lacks clarity	1

Skill Taxonomy Structure

- **Data Loading & Management**
 - Loading dictionaries/lists
 - Adding new entries
 - Structuring data properly
- **Data Cleaning**
 - Lowercasing, punctuation removal
 - Space normalization
 - Slang/keyword corrections
 - Consistency checks
- **Functions & Analysis**
 - User-defined functions
 - Loops, conditionals
 - Word counting, logical operations
 - Basic statistical measures
- **Data Summarization**
 - Sorting data
 - Filtering and grouping
 - Extracting unique words

Evaluation Guidelines

- Review the complete code, including code, outputs, and explanations.
- For each section, match student work to the specific skill and sub-skill area in the taxonomy.
- Assign full or partial marks per the rubric, using suggested partial mark ranges for “partially complete” work.
- For ambiguous or incomplete submissions, note what’s missing and correspondingly reduce credit.
- Comments must explain strengths and improvement points for each section.
- Objective assessment: focus on function, correctness, clarity, not on visual design beyond the stated requirements.

Achievement Criteria & Passing Rules

- At least partial (not zero) credit must be received in every section—no component can be left unattempted.
- Students must fully achieve at least two major skills: **Data Cleaning & Text Processing, Functions, Loops & Logic and Summary Statistics(min,max,count..)**
- Feedback and a detailed scorecard must be given for all evaluated assignments.